ARC2-Global: Tangent-Corrected Alignment for Embedding-Index Migration

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Abstract. Embedding-model upgrades require legacy document vectors to remain searchable while a new representation is introduced. We present ARC2-Global, a document adapter that fits two closed-form, norm-scaled negative cosine-gradient targets while retaining a centered Procrustes initialization. Both corrections collapse into one affine map, allowing native destination queries to search mapped and newly encoded documents together. Across four development collections, two encoder migrations and three destination-document budgets, ARC2-Global improves macro nDCG@10 over all five evaluated baseline configurations. It exceeds Procrustes in 23 of 24 condition means, with a macro gain of 0.0243 and Bonferroni-adjusted interval [0.0181, 0.0308]. Ablations establish an additional 0.0069 development gain from the second correction. On prospective TREC-COVID/E5, gains over the evaluated Converter range from 0.2000 to 0.2572 across the three budgets. On MiniLM, 75% coverage gives 0.5195 nDCG versus native Full-Y's 0.5203. The final 384-to-768-dimensional adapter occupies 1.13 MiB, uses 114.7 times fewer parameter bytes than that Converter, and requires no neural mapper training.

Keywords: Embedding migration · Vector retrieval · Representation alignment · Mixed galleries

1 Introduction

Embedding-model upgrades create a mismatch between destination-model queries and legacy document vectors. We study transitions where destination-space queries are required while document encoding proceeds gradually. A staged solution learns from paired source and destination embeddings, maps the remaining document vectors, and replaces those proxies as true destination vectors become available. The data-management question is the quality of the index at intermediate migration budgets.

The intermediate index must rank true destination embeddings and mapped proxies together. A mapper that works well on a proxy-only gallery can change behavior after true vectors are inserted. Cosine retrieval also depends on direction, whereas coordinate regression spends capacity on radial errors that disappear after normalization. This motivates a directional document adapter that preserves compact inference and searches with the native destination query.

Document-side conversion and query-side adaptation already provide routes across embedding spaces [17,11]. Orthogonal alignment supplies a compact geometric starting point, while local alignment and neural regression provide more flexible alternatives [16,2]. We investigate a complementary design: correcting the prediction error that changes vector direction while retaining one affine document map.

ARC2-Global applies two tangent corrections to a centered rectangular Procrustes map. For each paired document, it projects destination error onto the tangent plane at the predicted direction, rescales it to the raw prediction, and fits an affine ridge update. Repeating this operation twice yields coefficients that collapse into one affine map followed by normalization. Native destination queries then search true and proxy vectors together, with each newly encoded document replacing its proxy under the same identifier (Fig. 1).

On four development collections, the fixed 1,024-pair adapter improves macro nDCG@10 over five baseline configurations across two encoder migrations and 25%, 50% and 75% destination coverage. The Procrustes gain is 0.0243; the second correction contributes 0.0069 beyond one step. Matched all-paid fitting supplies an additional MiniLM operating point. A prospective TREC-COVID evaluation confirms gains over selected neural and regression configurations, including E5 improvements at every budget against Converter and ridge.
Our contributions are:

- **Directional refinement with affine inference.** Two tangent-residual ridge updates improve normalized alignment while retaining a single compact document map and native destination queries.
- **Strong quality at matched fitting exposure.** Across two migrations and three paid-document budgets, the four-collection development macro favors ARC2-Global over all five evaluated alternatives. Ablations quantify the value of the second update, and all-paid fitting tests the method beyond fixed-anchor supervision.
- **Prospective transfer and practical migration insight.** A held-collection evaluation confirms E5 gains over Converter and affine ridge at every tested budget. Paired gallery diagnostics identify when true-vector replacement changes the ranking benefit of a fitted adapter.

2 Related Work

Document-side translation and mixed databases. Embedding-Converter learns document-side conversion using regression together with global and local similarity losses [17]. Yang et al. study integration across embedding models through global alignment and locally fitted isometries, giving the A2M and LA2M approaches [16]. Their authors' repository also provides the SimpleLinear neural mapper used in our comparison [2]. More recently, Yang and Cao study out-of-distribution translation, multi-model mixing, and chaining, with a hierarchical mixture of experts and residual regularization along target principal directions [15]. Their localized expert structure addresses broader multi-model

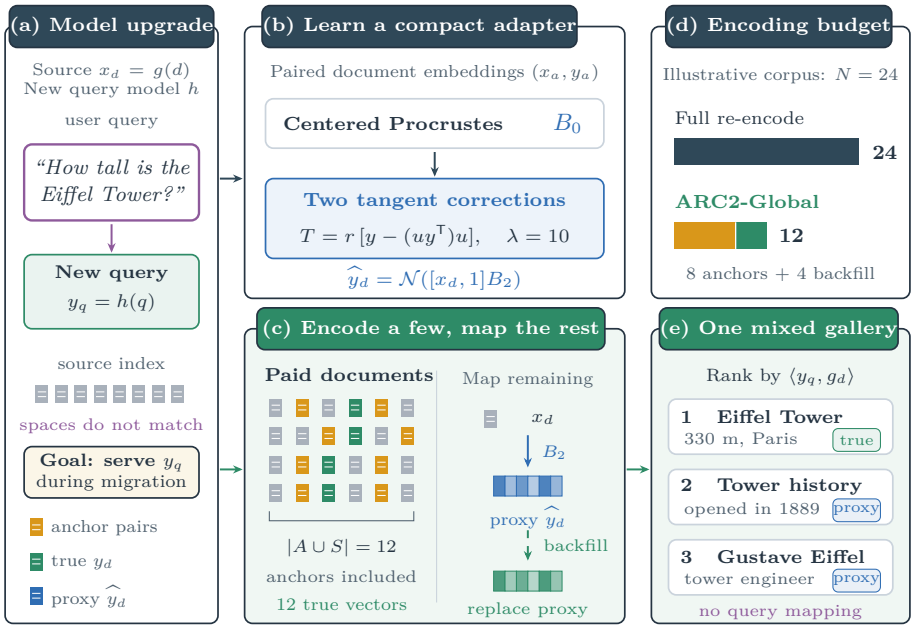


Fig. 1. A worked ARC2-Global migration. Paired document vectors train centered Procrustes and two tangent corrections, yielding one affine adapter. Unencoded documents remain searchable through proxies; backfill replaces them with native vectors in the same gallery. The 24-document corpus, eight anchors, four additional encodings and displayed hits are illustrative, not measured retrieval results.

integration; our directional target uses the tangent plane at each anchor's current prediction, and both corrections collapse into one global affine map. HMoE provides related-work context; our evaluated panel comprises the five configurations specified in Sect. 4. FastFill already combines visual feature alignment with uncertainty-based partial backfilling, and shows that replacement can initially reduce retrieval performance [3]. Building on these migration settings, we study the two-correction affine computation and its relevance-judged text-retrieval behavior under anchor-inclusive encoding budgets. Our comparison instantiates five alignment and regression configurations under common paired-document supervision (Sect. 4).

Compatibility and translation direction. Backward-compatible representation learning trains a new encoder to remain usable with previously computed gallery features [9]. Drift-Adapter instead fits a post-hoc map from new-model queries to an existing legacy index, comparing Procrustes, low-rank affine, and residual neural adapters [11]. Spherical interpolation refines Procrustes-aligned queries through SLERP between old and aligned new query directions [7]; ARC2 instead fits tangent ridge updates to document anchors. ARC2-Global shares the use of

paired embeddings and frozen encoders, but maps legacy document vectors into the destination space. Its evaluation concerns mapped and native destination documents competing as re-encoding progresses.

Geometric alignment and refinement. Orthogonal Procrustes provides the classical least-squares alignment underlying our initialization [8]. Recent work connects approximate preservation of pairwise inner products to bounds on Procrustes alignment error and demonstrates interoperability across retrieval models [5]. Unpaired translation pursues a different supervision regime: vec2vec learns translation through a shared latent representation without known correspondences [4], while mini-vec2vec combines pseudo-pair construction, Procrustes estimation, and iterative matching and clustering refinements [1]. Thus, linear translation, Procrustes initialization, and refinement are all established ingredients. ARC2-Global uses known document correspondences and a specific finite refinement: exactly two ridge-fitted tangent corrections to a centered rectangular Procrustes initialization. Both corrections are folded into one affine coefficient matrix, followed by row normalization at application time. This differs from discovering pseudo-correspondences, learning a latent translator, or routing documents among local maps.

Retrieval evaluation. BEIR emphasizes retrieval across heterogeneous tasks and domains [10], while NanoBEIR provides smaller collections suitable for development comparisons [18]. Sentence-BERT, MiniLM, and E5 supply established encoder families and training approaches relevant to our model transitions [6,14,13]. Our evaluation targets judged retrieval quality in actual mixed galleries, separating that quantity from agreement with a destination ranking or approximate nearest-neighbor overlap. This distinction matters because accurately reconstructing individual vectors and preserving their internal neighborhoods do not directly measure the relevance of the final combined ranking. The experimental question is therefore whether the correction improves retrieval under a matched encoding budget, and in which encoder and gallery conditions that improvement persists.

3 ARC2-Global: Angular Refinement for Mixed Galleries

ARC2-Global converts existing document embeddings into destination-space proxies, then combines these proxies with true destination document vectors. Its central design is to refine the directions that determine cosine scores while retaining a single global affine map. We first define the migration budget, then describe the initialization, two angular corrections, and resulting retrieval gallery.

3.1 Migration Setting and Budget

Let a corpus contain N documents with stored, unit-normalized source embeddings $x_i \in \mathbb{R}^{d_s}$, written as row vectors. The destination encoder produces unit

vectors $y_i \in \mathbb{R}^{d_t}$, but obtaining every y_i requires encoding the entire corpus again. We seek destination-space document representations using only a budgeted set of destination document vectors. The dimensions may differ; our evaluated encoder pairs map $d_s = 384$ to $d_t = 768$. Queries are encoded directly by the destination query encoder. The document map therefore has to align with an existing destination query space, rather than jointly redefining both sides of retrieval.

Let A be the paired anchor set used for fitting and S the set selected for document replacement. The unique destination-document budget is

$$b = |A \cup S|, \qquad b = \lfloor \rho N \rfloor. \tag{1}$$

Anchors consume this budget. Our selection starts with A and fills the remaining slots from a fixed random document order, skipping duplicates, so $A \subseteq S$ and $|S| = b$. Larger budgets extend the same order and retain the same fitted anchors. The principal setting uses $m = |A| = 1{,}024$ pairs; an all-paid-pairs comparison separately fits on every paid pair. Fit uses paired document vectors without queries or relevance labels. This separates the amount of fitting supervision from the number of true vectors in the gallery. In particular, increasing the replacement budget need not change the map: the same proxies can be evaluated at several stages of replacement. At a fully replaced gallery, document scores depend entirely on the destination encoder, irrespective of the fitted proxy map.

3.2 Centered Rectangular Initialization

A centered Procrustes map supplies a geometric starting point before angular refinement. Stack the source and destination anchors as $X_A \in \mathbb{R}^{m \times d_s}$ and $Y_A \in \mathbb{R}^{m \times d_t}$, with row means μ_s and μ_t. Using a thin singular value decomposition, compute

$$(Y_A - \mathbf{1}\mu_t)^\top (X_A - \mathbf{1}\mu_s) = U \Sigma V^\top, \qquad W = UV^\top. \tag{2}$$

Here $W \in \mathbb{R}^{d_t \times d_s}$ is rectangular and semiorthogonal. Centering permits translation between the two anchor clouds; the rectangular map accommodates their different dimensions. Its raw prediction is $(x - \mu_s)W^\top + \mu_t$.

We expand this prediction into an affine coefficient matrix so that subsequent corrections can be accumulated without adding inference stages. With $p = d_s + 1$ and $\widetilde{x} = [x, 1]$, define

$$B_0 = \begin{bmatrix} W^\top \\ \mu_t - \mu_s W^\top \end{bmatrix} \in \mathbb{R}^{p \times d_t}, \qquad \widetilde{x} B_0 = (x - \mu_s)W^\top + \mu_t. \tag{3}$$

This expansion preserves the raw centered prediction: it introduces neither row normalization nor a fitted scale. That distinction matters because the angular corrections below explicitly use the raw prediction norms. The raw affine expansion uses float64 coefficients computed from the float32 Procrustes means and matrix.

3.3 Two Tangent-Residual Ridge Corrections

Write $\mathcal{N}(z) = z/\|z\|_2$ for a nonzero vector z. Cosine retrieval depends on document direction, whereas a raw vector residual also contains a radial component. We therefore project each anchor target onto the tangent plane at its current predicted direction. Let $\widetilde{X} = [X_A, \mathbf{1}]$. At update $t \in \{0, 1\}$, for anchor j set

$$z_j^{(t)} = \widetilde{x}_j B_t, \qquad r_j^{(t)} = \|z_j^{(t)}\|_2, \qquad u_j^{(t)} = z_j^{(t)}/r_j^{(t)},$$
$$T_j^{(t)} = r_j^{(t)} \left[y_j - \left(u_j^{(t)} y_j^\top\right) u_j^{(t)} \right]. \tag{4}$$

The bracketed term is orthogonal to $u_j^{(t)}$. The norm factor converts this directional discrepancy into a target in raw prediction coordinates.

The target has a direct cosine-loss interpretation. For a nonzero row prediction z and unit destination anchor y_j, define $\ell_j(z) = 1 - y_j \mathcal{N}(z)^\top$. Differentiation gives

$$\nabla_z \ell_j(z) = -\frac{y_j - (u y_j^\top) u}{\|z\|_2}, \qquad T_j^{(t)} = -(r_j^{(t)})^2 \nabla_z \ell_j(z_j^{(t)}). \tag{5}$$

Thus the target is a norm-scaled negative cosine gradient, expressed in raw prediction coordinates. The shared affine regression smooths these targets across anchors; its fitted row corrections need not themselves be tangent. Retrieval benefit is assessed for the finite fitted procedure.

One global ridge regression shares these correction targets across anchors. Stacking their rows into $T^{(t)}$, we solve

$$\Delta_t = \underset{\Delta \in \mathbb{R}^{p \times d_t}}{\arg\min} \|\widetilde{X}\Delta - T^{(t)}\|_F^2 + \lambda \|\Delta\|_F^2, \tag{6}$$
$$= \widetilde{X}^\top (\widetilde{X}\widetilde{X}^\top + \lambda I_m)^{-1} T^{(t)}, \qquad B_{t+1} = B_t + \Delta_t. \tag{7}$$

The penalty regularizes the correction, including its intercept. Equivalently, changing variables to the updated coefficient matrix yields

$$B_{t+1} = \underset{C}{\arg\min} \|\widetilde{X}C - (\widetilde{X}B_t + T^{(t)})\|_F^2 + \lambda \|C - B_t\|_F^2. \tag{8}$$

This fit centers its correction penalty on the current affine map while matching norm-scaled cosine descent targets. Unlike direct zero-centered ridge onto Y_A, the update retains the existing alignment while fitting its directional correction. We fix $\lambda = 10$ and perform exactly two updates, recomputing targets after the first; the development ablation in Sect. 5 measures the second update's contribution. A Cholesky factorization is reused across updates. The refined map is unconstrained; fitting requires paired document vectors and linear solves, with no neural mapper training.

The norm factor also gives positive global scale equivariance. In exact arithmetic with nonzero predictions, $T(cB) = cT(B)$ for a common $c > 0$. Linearity of the ridge solve therefore scales each update by c; by induction both accumulated maps scale by c, leaving normalized inference unchanged. This property concerns a common coefficient scale, not arbitrary per-document changes or guaranteed finite-step cosine improvement.

Table 1. ARC2-Global fitting and gallery construction. All document vectors supplied to fitting or replacement are unit normalized.

1 Form paired anchors X_A, Y_A and augmented design $\widetilde{X} = [X_A, \mathbf{1}]$.
2 Compute centered Procrustes and raw affine B_0 using Eqs. (2)–(3).
3 Factor $\widetilde{X}\widetilde{X}^\top + 10I_m$ once.
4 For $t = 0, 1$: compute $T^{(t)}$ from Eq. (4), solve Eq. (7), and accumulate B_{t+1}.
5 Store B_2; map documents with $\widehat{y}_i = \mathcal{N}([x_i, 1]B_2)$.
6 Replace rows in S by y_i; search with destination query vectors.

3.4 Affine Proxies and Mixed Retrieval

The two corrections collapse into the final coefficient matrix B_2. For each document, compute a proxy $\widehat{y}_i = \mathcal{N}([x_i, 1]B_2)$ and construct

$$g_i = \begin{cases} y_i, & i \in S, \\ \widehat{y}_i, & i \notin S. \end{cases} \tag{9}$$

Every anchor is replaced by its actual fitted destination vector. For a unit destination query q, rank documents by $qg_i^\top$. True and proxy vectors therefore compete in one score space. This competition makes mixed-gallery quality a distinct evaluation target: proxy approximation quality alone does not determine how replacement changes their relative scores. We use exact float32 inner products, sorting by descending score and then ascending document row index for ties; dataset-defined identical query/document IDs are excluded where specified. Table 1 summarizes the complete computation.

3.5 Computation and Storage

The final mapper has pd_t coefficients regardless of anchor count. For $384 \to 768$, this is 295,680 float32 values, or 1,182,720 bytes (1.13 MiB), including the intercept. This is coefficient storage; fitting objects, document galleries, and encoder weights are additional. Application requires one affine multiplication and normalization per document, with $O(Npd_t)$ arithmetic for a complete proxy gallery.

Fitting cost depends on the solver backend. Beyond the cross-covariance and rectangular SVD, costing $O(md_sd_t)$ and $O(d_sd_t \min(d_s, d_t))$, respectively, the reference dual implementation forms an $m \times m$ Gram matrix in $O(m^2p)$ time, factors it in $O(m^3)$, and spends $O(m^2d_t + mpd_t)$ per correction. Its Gram storage is $O(m^2)$. A separately identified primal backend solves $(\widetilde{X}^\top\widetilde{X} + \lambda I_p)\Delta_t = \widetilde{X}^\top T^{(t)}$ instead, with $O(mp^2 + p^3)$ construction and factorization and $O(p^2)$ Gram storage; it supports larger paired sets. Each primal correction costs $O(mpd_t + p^2d_t)$. The equations are equivalent in exact arithmetic, while floating-point rankings can differ near ties. Updates use float64 and the final coefficient is stored as float32. Finally, $|A \cup S|$ measures logical destination-vector exposure in cached-vector evaluations. It does not measure physical encoding work, deployment latency, or end-to-end migration savings.

4 Experimental Setup

We ask three questions: does angular refinement improve fixed-supervision alignment, how does it compare with other mappers at a matched encoding budget, and which gains transfer to a prospectively held collection? All experiments use exact retrieval, isolating representation quality from approximate-index effects.

Collections and migrations. Development uses FiQA (57,638 documents, 500 development queries), NanoNQ (5,035/50), NanoHotpotQA (5,090/50), and NanoFEVER (4,996/50), drawn from BEIR and NanoBEIR [10,18]. FiQA informed method selection after its scores were inspected; all four collections are development evidence. We subsequently fixed a separate confirmation on the BEIR release of TREC-COVID [12,10]. Its input manifest contains all 171,332 documents and 50 queries. For this study, labels remained closed until every ranking and endpoint had been sealed and separately verified. This distinguishes development evidence from the prospective test. We evaluate two 384-to-768-dimensional migrations: `all-MiniLM-L6-v2` to `all-distilroberta-v1`, and `e5-small-v2` to `e5-base-v2`.[1] Model revisions and input hashes are pinned. Text encoding uses normalized float32 vectors, a maximum of 128 tokens, and the E5 query/passage prefixes.

Matched supervision and exposure. The main matrix crosses four development collections, two migrations, three paid-document budgets (25/50/75%), six methods, and seeds 17, 29, 41, 53, and 67. All methods fit the same ordered 1,024 paired anchors and replace the same paid document rows. Paid sets contain all anchors and are nested, with exactly $\lfloor \rho N \rfloor$ unique target document vectors; the accounting is $|A \cup S|$, including anchors. On the Nano collections and TREC-COVID, independent seeded streams sample anchors without replacement and a document permutation; unseen permutation rows fill the budget. FiQA retains its historical anchor/order arrays. Neither fitting nor selection uses relevance labels. Paid nonanchor vectors provide replacement only in this fixed-anchor regime. A separate FiQA experiment holds the 50% paid set fixed and compares fitting on 1,024 versus all 28,819 paid pairs.

Comparison methods. We retain the full five-baseline panel: centered rectangular Procrustes (an A2M adaptation); LA2M with 20 source-anchor clusters, local alignment for clusters with at least two anchors and global fallback for singletons; SimpleLinear; affine ridge with $\lambda = 10$ and a regularized bias; and EmbeddingConverter [16,2,17]. SimpleLinear uses pinned official code: despite its name, the locked configuration is a five-layer SELU MLP, trained for the final 50 epochs with batch size 4,028 and the MSE plus cosine loss, without qrel-based stopping. Converter is a paper-specification adaptation with a 50,000-update selection run, batch size 64, and local neighborhood size 100. Paired-vector validation MAE selects the earliest minimum at 250-update checkpoints; a fresh fit uses all fitting

[1] Pinned maintainer cards: MiniLM, DistilRoBERTa, E5-small, and E5-base.

pairs for that selected length. Unlike the original retrieval-based validation, this adaptation uses paired-vector MAE to respect paired-document-only supervision. These are the declared implementations and training policies, rather than claims to reproduce every original-paper setting. ARC2-Global remains fixed at two tangent updates and $\lambda = 10$ throughout.

Retrieval and inference. Destination queries search a single gallery containing unit-normalized proxies and paid true vectors, using exact float32 inner products and top-100 retrieval. Score ties use original document row order; identical query/document IDs are excluded. Native Full-X and Full-Y use their own query/document encoders throughout and provide reference operating points. We report graded nDCG@10 with gain $2^{\text{rel}} - 1$ and Recall@100 over all positive judgments, including positives outside a Nano corpus. All queries count; unjudged documents are nonrelevant.

Each locked analysis uses 20,000 paired query-bootstrap draws after averaging the five mapping seeds. Development macro effects weight the 24 collection/migration/budget conditions equally and resample queries within each collection. Within each analysis, the same sampled query indices are shared across methods, budgets and migrations for that collection. Saved 99% percentile intervals give approximate 95% coverage for the five-comparison family; the ablations have a separate five-contrast family. The all-paid experiment uses its predeclared 32-contrast family. TREC-COVID has 30 primary contrasts (five baselines, two migrations, three budgets), with approximately 99.8333% individual percentile intervals for approximate 95% family coverage. Its independent units are the 50 queries. Recall intervals are nominal 95% descriptive intervals. An interval touching zero is unresolved. These intervals condition on the five observed mapping seeds and fixed collections and encoder pairs; development intervals do not adjust for method selection.[2]

5 Results

5.1 Broad Retrieval Gains with a Compact Adapter

ARC2-Global delivers the highest macro nDCG@10 among all six evaluated methods at matched fitting exposure (Table 2). Against the closest comparator, Procrustes, the gain is 0.024325, with family interval $[0.018132, 0.030792]$. The gain appears in both migrations: the descriptive pair-wise macro differences are 0.025733 for MiniLM and 0.022917 for E5. Across individual conditions, ARC2 has higher nDCG point means in 23 of 24 comparisons with Procrustes and all 24 comparisons with each other baseline. The breadth of the point gains complements the family-adjusted macro comparison. Figure 2 shows the complete budget curves.

[2] Code, full baseline settings (`configs/baselines.json`) and manifests are packaged in a separate research reproducibility artifact. An anonymous public link is pending.

Table 2. Fixed-anchor retrieval quality, equally weighted over four development collections, two migrations and three budgets. All methods fit the same 1,024 pairs. Differences are ARC2 minus baseline; 99% individual intervals give approximate 95% coverage for the five-comparison family.

Method	nDCG@10	R@100	Δ nDCG	Family interval
ARC2-Global	0.5275	0.8659	—	—
Procrustes	0.5032	0.8661	+0.0243	[+0.0181, +0.0308]
LA2M20	0.4234	0.7758	+0.1041	[+0.0861, +0.1229]
SimpleLinear	0.3554	0.6130	+0.1721	[+0.1499, +0.1947]
Affine ridge	0.4318	0.7441	+0.0957	[+0.0754, +0.1164]
Converter	0.3663	0.7641	+0.1612	[+0.1427, +0.1799]

The advantage is concentrated in the top-ranked results: macro Recall is 0.8659, versus Procrustes' 0.8661, with a descriptive difference interval $[-0.004944, 0.004336]$. The exception to the 23-of-24 nDCG point gains is FiQA/E5 at 75% (-0.003583). Native Full-X and Full-Y remain visible as full-corpus reference operating points.

5.2 The Second Correction Adds Measurable Value

The second update provides a measurable benefit: two versus one step improves macro nDCG by 0.006924, with family interval $[0.003809, 0.009959]$. The gain over affine ridge is 0.095653 for the complete initialized refinement procedure in this supervision regime (Fig. 3); this contrast evaluates tangent targeting and the retained alignment prior together. The penalty controls measure sensitivity around the fixed setting. Changing λ from 10 to 1 gives +0.003197, with an interval crossing zero; changing it to 100 gives -0.009255, with a wholly negative interval. We retain the frozen setting.

5.3 Prospective Gains Transfer to TREC-COVID

The held collection confirms a substantial E5 advantage over the evaluated Converter and affine ridge at **all three migration budgets**. At 25% destination coverage, the gain over Converter is 0.257234, with family interval $[0.175731, 0.337569]$. At 50%, ARC2 reaches 0.5904 nDCG, versus 0.3590 for Converter and 0.3281 for ridge (Table 3). Across both migrations, family-adjusted improvements occur in five of six Converter comparisons, four ridge comparisons and two SimpleLinear comparisons: **11 improvements in the 30-contrast family**. Table 3 marks every supported pairwise improvement and retains all six methods at all three budgets. Figure 4 shows how the gains change with coverage.

The MiniLM migration supplies a useful partial-coverage operating point: at 75% coverage, ARC2-Global reaches 0.5195 nDCG against native Full-Y's 0.5203. These point estimates illustrate proximity to destination-model quality

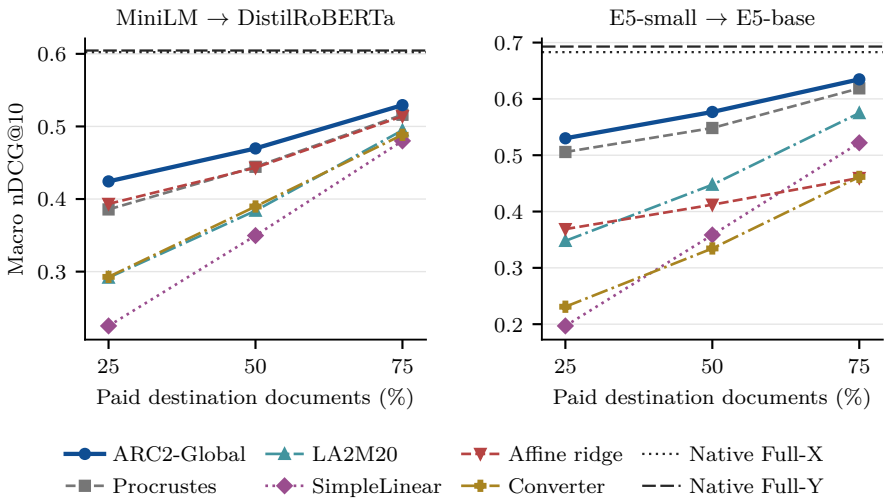


Fig. 2. Development quality versus anchor-inclusive paid-document coverage. Each point equally averages the four collection means after averaging five mapping seeds. Native references use the same collection weighting. Curves are descriptive; no pooled uncertainty band is implied.

while one quarter of the gallery still uses proxies. The predeclared inferential family compares adapters; it does not establish equivalence to native Full-Y.

The strongest geometric alternatives remain competitive: Procrustes has higher point means in all six cells, while the Procrustes and LA2M family intervals cross zero. Thus the prospective advantage is specific to the evaluated Converter, ridge and SimpleLinear configurations. The native references in Table 3 retain the quality context for cutover.

5.4 Using Paid Pairs Strengthens MiniLM Adaptation

More fitting supervision produces a strong MiniLM operating point without increasing the paid-document set. With all paid FiQA pairs, ARC2 reaches 0.367146 nDCG and exceeds Procrustes, LA2M, SimpleLinear, and Converter in the locked family comparisons; ridge remains unresolved. E5 shows a different ordering: ARC2 trails Procrustes and LA2M, exceeds SimpleLinear and ridge, and is unresolved against Converter. Table 4 retains both migrations and all methods. Fixed-anchor MiniLM ARC2 trails each all-paid baseline in the locked cross-regime nDCG comparisons. Increasing ARC2's fitting exposure raises MiniLM nDCG from 0.307505 to 0.367146; E5 changes from 0.311635 to 0.291867, with that nDCG interval unresolved. All-paid fitting retains $\lambda = 10$ in the summed objective, so it evaluates the fixed procedure at larger supervision rather than isolating data quantity at a fixed effective penalty. The result identifies when the

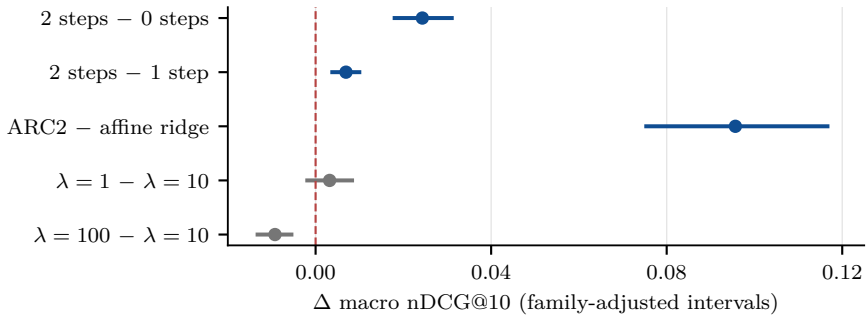


Fig. 3. Development mechanism controls. Dots are equal-condition macro nDCG differences; bars are the saved 99% intervals of the separate five-contrast family. The final two rows vary the penalty while retaining two steps.

Table 3. Prospective TREC-COVID nDCG@10, averaged over five mapping seeds. A baseline cell marked † has a positive ARC2-minus-baseline interval in the predeclared 30-contrast family (approximate 95% family coverage). Unmarked cells have intervals crossing zero. Full-X/Y are native full-corpus references.

Method	MiniLM → DistilRoBERTa			E5-small → E5-base		
	25%	50%	75%	25%	50%	75%
ARC2-Global	0.4619	0.5000	0.5195	0.5628	0.5904	0.6098
Procrustes	0.4620	0.5046	0.5195	0.5660	0.6067	0.6198
LA2M20	0.4634	0.5073	0.5186	0.5206	0.5898	0.6188
SimpleLinear	0.4610	0.5076	0.5193	0.4299†	0.5287†	0.5785
Affine ridge	0.4322	0.4854	0.5012†	0.2986†	0.3281†	0.3278†
Converter	0.4224†	0.4780†	0.5033	0.3056†	0.3590†	0.4098†
Native Full-X	0.4935 (native)			0.6805 (native)		
Native Full-Y	0.5203 (native)			0.6421 (native)		

paid pairs can improve adaptation, while the two-migration comparison captures the change in method ordering.

A practical insight: validate the mixed gallery. Holding each fitted mapper fixed, we compare proxy-only and mixed galleries under the same document replacement sets. For all-paid E5 ARC2 on FiQA, nDCG decreases from 0.314908 to 0.291867; Recall decreases from 0.654164 to 0.559490. The paid-document share of top-10 results falls from 0.491640 to 0.064720 despite 50% paid coverage. For MiniLM ARC2, nDCG instead rises from 0.350951 to 0.367146. These opposite replacement effects make mixed-gallery relevance the decision metric for partial migration. Good proxy-only quality is insufficient to select a cutover point; the documents must be ranked together in the intended gallery.

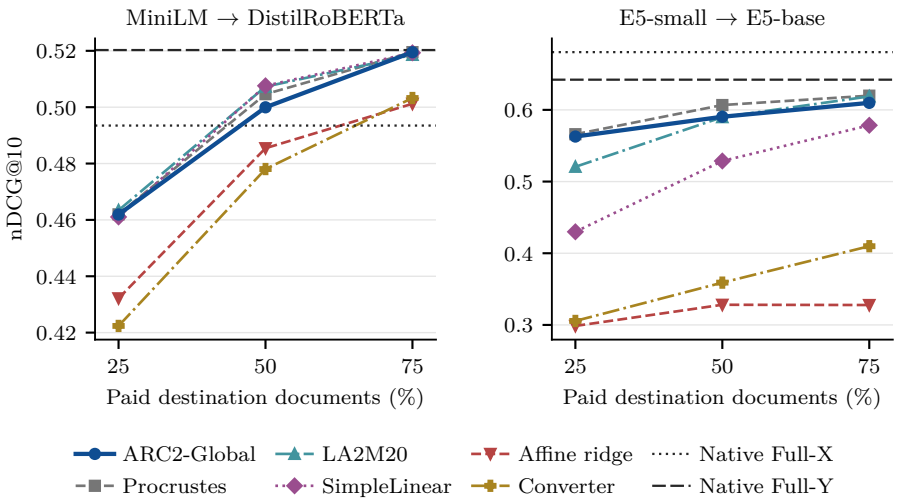


Fig. 4. Prospective TREC-COVID quality at fixed 1,024-anchor supervision. All six methods and native references are shown; points average the five mapping seeds. The two panels use different vertical ranges.

6 Practical Value of Directional Adaptation

Better ranking without a larger inference model. ARC2-Global combines the strongest development macro quality in the evaluated panel with a compact deployment artifact. For 384-to-768-dimensional conversion, its 295,680 float32 coefficients occupy 1.13 MiB: 114.7 times fewer parameter bytes than the evaluated 33,927,168-parameter Converter. The two corrections are absorbed into the affine coefficients, so they add no stages to document mapping at inference. Affine ridge has the same coefficient count; the comparison therefore separates the retrieval benefit of the correction procedure from the storage benefit of affine mapping relative to the evaluated neural alternative.

A clear operating scenario. The intended user must support destination-model queries while retaining a single destination-space gallery during staged document replacement. The fixed-anchor study demonstrates strong mapping quality with limited paired supervision; the all-paid MiniLM result shows a useful operating point when all available paid pairs are used. Prospective E5 gains over Converter and ridge persist through 25%, 50% and 75% coverage. These gains complement the TREC-COVID/MiniLM operating point: 0.5195 nDCG at 75% coverage versus 0.5203 for native Full-Y. These measured conditions make ARC2-Global a concrete candidate for evaluating a required encoder transition. If source queries can remain in service, native Full-X remains a viable alternative: every partial development ARC2 endpoint is below it. Procrustes and LA2M also remain

Table 4. FiQA at the same 50% paid-document budget. Fixed uses 1,024 fitting pairs; All uses all 28,819 paid pairs. N denotes nDCG@10 and R denotes Recall@100. Native references use their own query/document encoders.

Method	MiniLM → DistilRoBERTa			E5-small → E5-base		
	Fixed N	All N	All R	Fixed N	All N	All R
ARC2-Global	0.3075	0.3671	0.6922	0.3116	0.2919	0.5595
Procrustes	0.2858	0.3334	0.6905	0.3100	0.3362	0.6534
LA2M20	0.2377	0.3389	0.6950	0.2300	0.3293	0.6473
SimpleLinear	0.2360	0.3323	0.6778	0.1942	0.1679	0.3618
Affine ridge	0.2890	0.3673	0.6800	0.1479	0.2270	0.4245
Converter	0.2496	0.3258	0.6633	0.1397	0.2625	0.5789
Native Full-X	0.3900 (native)		0.7002	0.3465 (native)		0.6453
Native Full-Y	0.4047 (native)		0.7039	0.3891 (native)		0.6968

strong choices on the prospective collection, so encoder-specific mixed-gallery validation should determine which adapter to deploy.

What the efficiency evidence measures. The compact final map and absence of neural mapper training are structural advantages. Our cached-vector experiments measure retrieval under matched destination exposure. Physical encoding time, fitting workspace, index resources and serving SLOs are separate deployment quantities; the parameter-storage ratio is not an end-to-end cost ratio.

7 Conclusion

ARC2-Global improves a compact document adapter by fitting two tangent residual corrections to a Procrustes initialization. The updates preserve affine inference while targeting prediction direction for normalized retrieval. Broad development comparisons, mechanism ablations, matched all-paid fitting and a prospective collection identify useful retrieval gains and encoder-dependent transfer. Together with the mixed-gallery diagnostics, these results support directional correction as a practical adapter design and mixed-endpoint evaluation as a necessary part of staged embedding migration.

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